

Adding and Subtracting Mixed Numbers

Answer Key

Grades 4–5 Number and Operations—Fractions

Quick Answers

$$\begin{array}{l} 1. \frac{7}{2} \\ 2. \frac{9}{2} \\ 3. \frac{9}{2} \\ 4. \frac{5}{2} \end{array}$$

$$\begin{array}{l} 5. 8 \\ 6. \frac{23}{6} \\ 7. \frac{37}{12} \\ 8. \frac{10}{3} \end{array}$$

$$\begin{array}{l} 9. \frac{7}{2} \\ 10. \frac{19}{12} \end{array}$$

Curriculum

Level: Grades 4–5 Number and Operations—Fractions

4.NF.B.3 – problems 1, 2, 3, 4, 5, 8, 9

5.NF.A.1 – problems 6, 7, 10

Difficulty 1–4 of 5 across 10 tagged checks.

1. Rosa’s walk: $2\frac{3}{8} + 1\frac{1}{8}$ miles.

Step 1. Both distances are measured in eighths, so the fraction parts can be added directly — no rewriting is needed.

Step 2. Wholes: $2 + 1 = 3$. Eighths: $\frac{3}{8} + \frac{1}{8} = \frac{4}{8}$.

Step 3. $\frac{4}{8}$ simplifies to $\frac{1}{2}$, so Rosa walked

$$3\frac{1}{2} \text{ miles}$$

Teaching note: the sum of the fraction parts is less than one whole here, so nothing is traded. Problem 3 is the case where something is.

2. Juice: $1\frac{5}{12} + 3\frac{1}{12}$ cups.

Step 1. Twelfths and twelfths again — like denominators, so add the two parts separately.

Step 2. Wholes: $1 + 3 = 4$. Twelfths: $\frac{5}{12} + \frac{1}{12} = \frac{6}{12}$.

Step 3. $\frac{6}{12}$ is one half, so the recipe uses

$$4\frac{1}{2} \text{ cups}$$

Listen for: an answer of $4\frac{6}{24}$ — the denominators were added too. Twelfths plus twelfths are still twelfths.

3. Practice time: $2\frac{5}{6} + 1\frac{4}{6}$ hours.

Step 1. Like denominators again, but watch the fraction column: five sixths plus four sixths is more than one whole, so this sum will need regrouping.

Step 2. Wholes: $2 + 1 = 3$. Sixths: $\frac{5}{6} + \frac{4}{6} = \frac{9}{6}$, which is $1\frac{3}{6}$.

Step 3. Trade the extra whole: $3 + 1\frac{3}{6} = 4\frac{3}{6}$, and $\frac{3}{6}$ is one half. Devon practised

$$4\frac{1}{2} \text{ hours}$$

Common wrong answer: $3\frac{9}{12}$ — the denominators were added along with the numerators.

4. Birdseed: $5\frac{1}{4} - 2\frac{3}{4}$ pounds.

Step 1. Three quarters cannot be taken from one quarter, so regroup before subtracting: borrow one whole from the 5.

Step 2. $5\frac{1}{4} = 4 + \frac{4}{4} + \frac{1}{4} = 4\frac{5}{4}$.

Step 3. Now subtract in columns: $4 - 2 = 2$ and $\frac{5}{4} - \frac{3}{4} = \frac{2}{4} = \frac{1}{2}$, so the birdseed left in the bag is

$$2\frac{1}{2} \text{ pounds}$$

Common wrong answer: $3\frac{2}{4}$ — the smaller fraction was taken from the larger one in the fraction column, which quietly changes the problem.

5. Pitcher: poured out $3\frac{2}{5}$ cups, $4\frac{3}{5}$ cups left.

Step 1. The starting amount is unknown, so name it x and write what the story does to it: $x - 3\frac{2}{5} = 4\frac{3}{5}$.

Step 2. Undo the pouring by adding the two known amounts: $x = 4\frac{3}{5} + 3\frac{2}{5}$.

Step 3. Wholes: $4 + 3 = 7$. Fifths: $\frac{3}{5} + \frac{2}{5} = \frac{5}{5}$, one more whole. So the pitcher held

$$8 \text{ cups}$$

Teaching note: the fraction parts made exactly one whole, so the answer has no fraction left. That is worth pointing out — it is the same trade as problem 3, just complete.

6. Broth: $1\frac{1}{2} + 2\frac{1}{3}$ cups.

Step 1. Halves and thirds are different-sized pieces, so they cannot be added until both are written over the same denominator. The smallest number both 2 and 3 divide is 6.

Step 2. $\frac{1}{2} = \frac{3}{6}$ and $\frac{1}{3} = \frac{2}{6}$.

Step 3. Wholes: $1 + 2 = 3$. Sixths: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$. The pot holds

$$3\frac{5}{6} \text{ cups}$$

Listen for: $3\frac{2}{5}$, which is what adding straight across gives. Ask which is bigger, a half or a fifth.

7. Bike path: $4\frac{3}{4} - 1\frac{2}{3}$ miles.

Step 1. Fourths and thirds again need one common denominator; 12 is the smallest number both 4 and 3 divide.

Step 2. $\frac{3}{4} = \frac{9}{12}$ and $\frac{2}{3} = \frac{8}{12}$. Nine twelfths is larger than eight twelfths, so no regrouping is needed.

Step 3. Wholes: $4 - 1 = 3$. Twelfths: $\frac{9}{12} - \frac{8}{12} = \frac{1}{12}$, so Priya still has to ride

$$3\frac{1}{12} \text{ miles}$$

Common wrong answer: $3\frac{1}{1}$, from subtracting numerators and denominators straight across. Denominators name the piece size; they are never subtracted.

8. Rain barrel: held $7\frac{1}{6}$ gallons, $3\frac{5}{6}$ gallons left.

Step 1. The amount used is unknown, so call it x and write the story as an equation: $7\frac{1}{6} - x = 3\frac{5}{6}$.

Step 2. That makes $x = 7\frac{1}{6} - 3\frac{5}{6}$. Five sixths cannot be taken from one sixth, so regroup: $7\frac{1}{6} = 6\frac{7}{6}$.

Step 3. Wholes: $6 - 3 = 3$. Sixths: $\frac{7}{6} - \frac{5}{6} = \frac{2}{6} = \frac{1}{3}$. The gardener used

$$3\frac{1}{3} \text{ gallons}$$

Teaching note: checking is quick and worth doing: $3\frac{1}{3} + 3\frac{5}{6}$ must return $7\frac{1}{6}$.

9. Error analysis: Sam wrote $4\frac{4}{8}$ for $6\frac{1}{8} - 2\frac{5}{8}$ inches.

Step 1. Look only at the fraction column. Sam needed $\frac{1}{8} - \frac{5}{8}$, could not do it, and turned it around into $\frac{5}{8} - \frac{1}{8}$ instead. Reversing a subtraction changes the problem; the fix is to regroup.

Step 2. Borrow one whole from the 6: $6\frac{1}{8} = 5\frac{9}{8}$ because one whole is $\frac{8}{8}$.

Step 3. Wholes: $5 - 2 = 3$. Eighths: $\frac{9}{8} - \frac{5}{8} = \frac{4}{8} = \frac{1}{2}$. The trim is

$3\frac{1}{2}$ inches

What to say: Sam's answer is bigger than the starting length minus two whole inches, which is a signal by itself. Estimating first ($6 - 3 \approx 3$) catches this error every time.

10. Flour: $3\frac{1}{4} - 1\frac{2}{3}$ cups.

Step 1. This one needs both moves: a common denominator first, and then regrouping, because a quarter is smaller than two thirds. Use twelfths.

Step 2. $3\frac{1}{4} = 3\frac{3}{12}$ and $1\frac{2}{3} = 1\frac{8}{12}$. Three twelfths is less than eight twelfths, so regroup: $3\frac{3}{12} = 2\frac{15}{12}$.

Step 3. Wholes: $2 - 1 = 1$. Twelfths: $\frac{15}{12} - \frac{8}{12} = \frac{7}{12}$, so Ana still needs

$1\frac{7}{12}$ cups

Teaching note: estimate first — a bit over 3 minus a bit under 2 should land near $1\frac{1}{2}$, and $1\frac{7}{12}$ does.